



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Term Exam Trimester: Summer 2025 Marks: 30 Time: 1 hr 30 mins

Code: CSE 3411 Course Title: System Analysis & Design

“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”

Answer the following questions:

1. Consider the following scenario and write down the answers to the questions mentioned as follows:

UIU plans to automate the course grade change and Incomplete (I) grade application processes through a new module. This module will allow students to submit applications for a grade change or an I grade along with a valid justification. Each application will be reviewed by both the course teacher and the program office, and either party may terminate the application with proper justification.

If the application is approved by both parties, the next steps depend on the type of application. For I grade applications, the course teacher schedules a make-up final exam and informs the student. Once the student completes the exam, the course teacher evaluates the script, scans it, prepares a revised grade sheet, and submits all necessary documents through the program office portal. For grade change applications, the course teacher directly uploads the updated grade sheet together with all supporting documents to the program office portal.

The program office reviews the submission and verifies it against the Student Result Data Table in the database. If any mismatch or error is detected, the course teacher is notified. If no issues are found, the application is forwarded to the Department Head and Dean's Office. At this stage, the student's result history table and the course teacher's performance record table may be checked for decision-making, and the application may still be rejected. If accepted, the application proceeds to the Vice Chancellor's office, which makes the final decision. Once approved by the VC, the Result Data Table is permanently updated with the new grade, and all relevant people get notification of the final decision.

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|---|--------------|
| a. Draw the DFD for the above scenario. | [CO3]
[4] |
| b. Draw the activity diagram. | [2] |
| c. Draw the Use case diagram for the above scenario (show at least one extend/include relationship in the diagram). | [2] |
| d. Write down the descriptive form of one major use case shown in the use case diagram for the above scenario. | [4] |

2. Considering the above-described scenario was in its initial stage before launch. Briefly explain any **one tool** of information gathering that would be most beneficial in gathering the information needed to make this project. Mention the merits and demerits of the **tool you chose**, and explain why your choice is better by comparing with **two other** less suitable tools.

[CO2] [1+2+2 = 5]

3. Answer the following questions:

[CO1]

- a. Do ES and DSS function in the same way? If not, explain **two** key differences between them. [2]
- b. Mention the different phases usually covered in SDLC? Explain how skipping or poorly performing in the Requirement Gathering phase can affect the success of a software project. [1+2 = 3]

4. You are evaluating a new e-commerce platform investment. At year 0, you invest **\$20,000**. Each following year, the investment increases by **\$10,000 more than the previous year's investment**, and this continues for the next two years. In return, the project generates cash inflows: in year 2, the return is **half of the investment made in year 1**, in year 3, it is **equal to the investment made in year 2**, in year 4, it is **twice the investment made in year 3**, and in year 5, it gives a return of **\$50,000**. The annual depreciation rate is **8%**.

[CO4]

Using the Cash Flow Method, determine whether this project results in a profit or loss.

[3]

5. Develop and present in the class room on a **partial** SRS document (software requirement specification) as a group wise project assigned in the classroom. The SRS document should cover the following areas regarding your respective project: Introduction, Rationale/purposes of the project, Detail system study (tentatively you should cover benchmark products, research papers, onsite visit, survey, online sources etc.), feasibility analysis (SWOT analysis, cash flow analysis etc.), system design (Use case diagram, use case descriptive form, Data Flow Diagram, activity diagram, swimlane diagram etc.)

(do not answer this question in the examination)

[CO3]

[5]